

Marketplace Fulfillment Risk Analysis

Executive Summary · Operations Analytics · Brazilian E-Commerce Dataset (100K orders, 2016–2018)

7.7x

More 1-star reviews from late deliveries

408 hrs

Extra transit time for late orders

0.813

Model AUC (time-based split)

50%

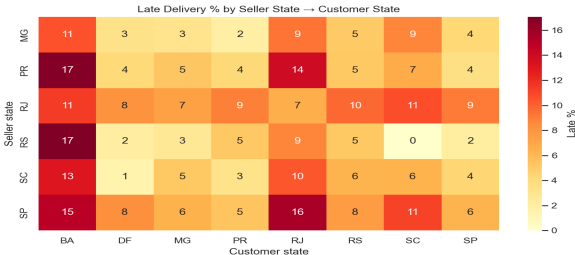
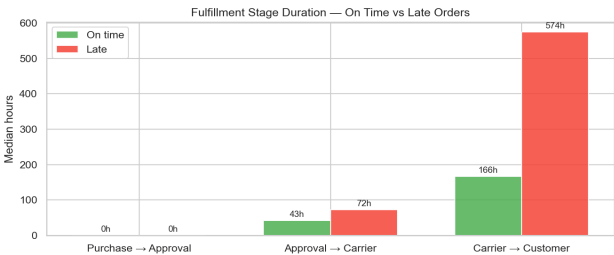
Late orders caught by flagging
13%

THE PROBLEM

Late deliveries are the leading driver of negative reviews. 46% of late orders receive a 1-star review compared to 6% for on-time orders. This analysis decomposes the fulfillment pipeline to locate the bottleneck, identifies which sellers and corridors are most at risk, and builds a predictive model that flags at-risk orders at placement time — enabling proactive intervention before the customer is affected.

KEY FINDINGS

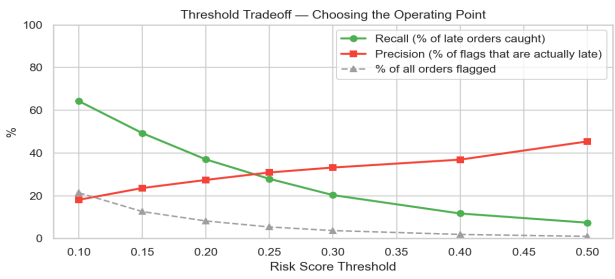
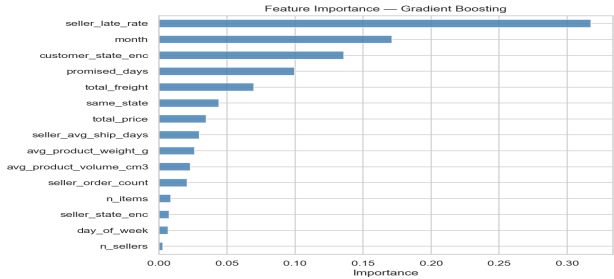
1. The transit leg causes 90%+ of the delay. Purchase-to-approval is near-instant. Seller prep adds a modest 29 extra hours for late orders. But carrier-to-customer delivery explodes from 166 hours (on-time) to 574 hours (late) — a gap of 17 days. The problem is in logistics and geography, not seller processing speed.



- 2. A small seller tail drives disproportionate lateness.** ~830 of 1,750 active sellers are Reliable ($\leq 5\%$ late). Only ~55 sellers (~3%) are At Risk ($>30\%$ late), yet they produce an outsized share of complaints. Median seller late rate is just 6%.
- 3. Geography is a structural risk factor.** Deliveries to BA and RJ consistently show 13–17% late rates regardless of seller origin. SP→RJ hits 16%; PR→BA and RS→BA hit 17%. Distance from the seller concentration in São Paulo is the underlying driver.

RISK-SCORING MODEL

A Gradient Boosting classifier predicts late delivery using only features available at order time (no data leakage). Trained on pre-April 2018 data, tested on April 2018+. AUC = 0.813. The seller's historical late rate alone accounts for 32% of predictive power, followed by month (17%), customer state (13%), and promised delivery window (9%).



RECOMMENDATIONS

- Adjust delivery estimates on flagged orders.** Add a 3–5 day buffer to promised windows for high-risk orders. Zero cost. Turns "late" into "on-time" by managing expectations upfront.
- Proactive customer notifications.** Auto-message customers on flagged orders if the package hasn't reached the carrier by day 7. Minimal cost. Reduces surprise and the resulting 1-star reviews.
- Seller accountability program.** Surface D-tier sellers ($>30\%$ late) in dashboards. Require improvement plans or reduce routing. Addresses the root cause.